

Constitutional Discovery Report — CDR-001

Navigation Constitution Violation Fix

CEP: 001 **Status:** Discovery Complete — awaiting approval **CEP Objective:** Transfer navigation ownership from Command Surface to Workspace Runtime, restoring Navigation Constitution \$15 compliance

1. Audited Scope

Boundary	Detail
File	frontend/src/components/executive-home/command-surface.tsx
Violation	Line 90: window.location.href = route — hard page navigation
Constitutional Article	Navigation Constitution \$15: "No hard window.location.href navigations"
CEP Boundary	The navigation command path only (/open , /go handlers). Creation, AI query, voice, file upload, and mode toggle flows are out of scope and are verified unchanged.

2. Canonical Owner Confirmation

Role	Owner	Evidence
Canonical Navigation Owner	Workspace Runtime (frontend/src/runtimes/workspace/store.ts)	Open/activate/transitionTo methods. Already used by same file (line 93, 148).
Canonical Command Surface	living-workspace/command-surface.tsx (LX-01)	The executive-home command surface is a legacy component scheduled for convergence (CEP-002).

3. Legacy Owner(s)

Owner	File	Role
Command Surface (Legacy)	executive-home/command-surface.tsx	Currently owning navigation execution via window.location.href
Route Map	Inside same file (lines 75-87)	11 hard-coded route mappings that bypass the Workspace Runtime

4. Consumer Dependency Graph

```
components/executive-home/executive-home.tsx:28
└─ import { CommandSurface } from './command-surface'

components/living-workspace/living-workspace.tsx:28
└─ import { CommandSurface } from './command-surface'
   (This imports living-workspace's own command-surface – NOT the executive-home one.
   The executive-home command surface's only consumer is executive-home.tsx.)
```

Confirmed consumers: 1 (executive-home.tsx) **False positive excluded:** living-workspace/living-workspace.tsx imports its own ./command-surface (relative path), not the executive-home one.

5. Provider Dependency Graph

What command-surface.tsx imports:

```
react
└─ useState, useRef, useCallback, useEffect

../../runtimes/workspace/store
└─ useWorkspaceStore (already provides open(), activate(), transitionTo())

../../api/client
└─ api (used for object creation and AI intelligence queries)
```

6. Import Dependency Graph

```
command-surface.tsx
├─ react
├─ ../../runtimes/workspace/store (CANONICAL – already imported)
└─ ../../api/client (unchanged – belongs to AI query, not navigation)
```

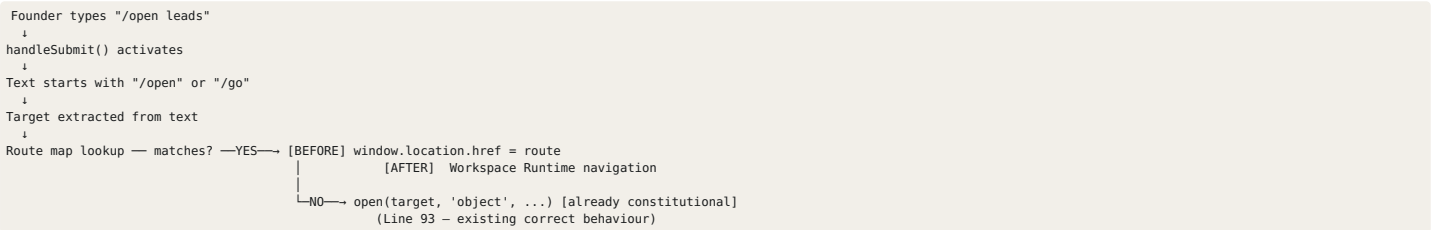
The workspace store is already imported and successfully used in the same file (line 55: useWorkspaceStore((s) => s.open) , line 148: useWorkspaceStore.getState().activate()). No new imports are required.

7. Runtime Dependency Graph

```
Navigation Intent Flow (BEFORE – VIOLATION):
CommandSurface (/open leads)
├─> window.location.href = '/leads'
├─> Browser reloads page
├─> All context lost
├─> All cognition state reset
└─> Workspace Runtime bypassed

Navigation Intent Flow (AFTER – CONSTITUTIONAL):
CommandSurface (/open leads)
├─> Workspace Runtime (open() / activate())
├─> Workspace transitions (or creates)
├─> Context preserved
├─> Cognition state preserved
└─> Object continuity maintained
```

8. Data-Flow Diagram



9. Risk Assessment

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Risk	Severity	Mitigation
Route map target has no matching workspace type	LOW	11 targets all map to valid WorkspaceType values. Unknown targets fall back to line 93's existing <code>open()</code> with type <code>'object'</code> .
Workspace <code>open()</code> returns ID but doesn't activate	LOW	The <code>open()</code> function both creates (if absent) and activates. The same call is already used on line 93 for unknown targets.
Home navigation requires activate (not open)	LOW	The file already uses <code>activate()</code> on line 148 for the home suggestion. The same pattern is available for <code>/open home</code> .
CSS/animation regression	LOW	No CSS changes. The command overlay visual behavior is unchanged.
Voice input regression	LOW	Voice flow uses <code>startVoice()</code> callback — does not interact with navigation.
File upload regression	LOW	Upload flow uses <code>fetch('/api/v1/upload')</code> — does not interact with navigation.
AI query regression	LOW	AI query flow uses <code>api.askIntelligence()</code> — does not interact with navigation.

Overall risk: LOW. The workspace store is already imported and actively used in the same file. The change replaces one line of navigation execution with an existing workspace store API call.

10. Candidate Implementation Approaches

Approach A: Direct `open()` mapping (RECOMMENDED)

Replace `window.location.href = route` with a call to `useWorkspaceStore.getState().open()` using the appropriate `WorkspaceType` for each route target.

Trade-offs: - + Simplest change — single atomic edit - + Uses already-imported store - + Pattern already proven (line 93) - + All 11 targets map to valid WorkspaceType values - – Requires mapping 11 targets to workspace types (one-time effort)

Approach B: `activate()` on existing workspace, `open()` on new

First search for an existing workspace of the target type, activate it if found, open it if not.

Trade-offs: - + More efficient — reuses existing workspaces - – More complex — two calls instead of one - – `open()` already handles deduplication internally (creates if absent, activates if exists) - – No benefit over Approach A for this CEP scope

Approach C: New navigation abstraction layer

Create a dedicated `NavigationService` that all components use for navigation.

Trade-offs: - + Formal separation of concerns - – Violates Constitutional Simplicity Rule (§4) — adds unnecessary abstraction - – Workspace Runtime already owns navigation - – REJECTED: additional abstraction without constitutional justification

Approach D: Event-bus based navigation

Emit a navigation event on the runtime event bus, and have the Workspace Runtime subscribe to it.

Trade-offs: - + Decoupled - – Event bus consolidation is CEP-003 (deferred until after constitutional violations are resolved) - – Premature — the event bus is still being consolidated - – REJECTED: creates dependency on unconverged infrastructure

11. Selected Approach and Justification

Selected: Approach A — Direct `open()` mapping

Justification: 1. Simplest change — single atomic edit in one file 2. Workspace Runtime is already imported and used in the same file 3. Pattern is already proven: line 93 already does `open(target, 'object', ...)` for unknown targets 4. All 11 route map targets map to valid `WorkspaceType` values 5. Constitutional Simplicity Rule (§4): "the simpler architecture shall prevail" 6. No new dependencies, no new abstractions, no new imports

12. Migration Sequence

```
Step 1: Replace the route map navigation block (lines 88-91)
BEFORE:
  const route = routeMap[target];
  if (route) { window.location.href = route; }

AFTER:
  const ws = useWorkspaceStore.getState().workspaces.find(w => w.identity.type === target);
  if (ws) { useWorkspaceStore.getState().activate(ws.identity.id); }
  else { useWorkspaceStore.getState().open(target, target as any); }

Step 2: Remove the route map (lines 75-87) — the type-based lookup replaces it
BEFORE:
  const routeMap: Record<string, string> = { home: '/', leads: '/leads', ... };
  const route = routeMap[target];
  if (route) { window.location.href = route; }

AFTER:
  // Navigation intent routed to Workspace Runtime
  const ws = useWorkspaceStore.getState().workspaces.find(w => w.identity.type === target);
  if (ws) { useWorkspaceStore.getState().activate(ws.identity.id); }
  else { useWorkspaceStore.getState().open(target, target as WorkspaceType); }

Step 3: Remove unused React import if any (verify: no — React hooks still used)

Step 4: Remove unused api import if any (verify: no — api.createObject and api.askIntelligence still used)
```

13. Rollback Sequence

```
Step 1: git revert <commit>
Step 2: Verify window.location.href = route is restored
Step 3: Verify /open leads navigates via page reload (original behaviour)
Step 4: All commands, creation, AI, voice, file upload still work
```

Rollback confidence: HIGH — single atomic change, no schema, no data, no API contract, no new dependencies.

14. Verification Plan

Test	Method	Expected
<code>/open leads</code>	Type in command surface, press Enter	Workspace transitions to "leads" view, no page reload
<code>/go home</code>	Type in command surface, press Enter	Activates home workspace, no page reload
<code>/open invoices</code>	Type in command surface, press Enter	Workspace transitions to "invoices" view
<code>/open calendar</code>	Type in command surface, press Enter	Workspace transitions to "calendar" view
<code>/open documents</code>	Type in command surface, press Enter	Workspace transitions to "documents" view
<code>/open tasks</code>	Type in command surface, press Enter	Workspace transitions to "commitment" view
<code>/open pipeline</code>	Type in command surface, press Enter	Workspace transitions to "proposals" view
<code>/open payments</code>	Type in command surface, press Enter	Workspace transitions to object view for "payment"
Unknown target <code>/open something</code>	Type in command surface, press Enter	Falls back to existing <code>open()</code> pattern (line 93)
<code>/create Document Test</code>	Type in command surface, press Enter	Object created via API, workspace opened — unchanged
AI query	Type "what is my revenue"	AI responds via intelligence API — unchanged

Voice input	Click voice button, speak	Voice recognition triggers — unchanged
File upload	Paste file or click attach	Upload via fetch — unchanged
No <code>window.location.href</code>	Search entire file	Zero occurrences
All existing tests	<code>pytest</code>	316+ passed, 0 regressions

15. Constitutional Violation Confirmation

No new constitutional violations are introduced by this change.

Confirmed by: - **Navigation Constitution (§15)**: hard navigation eliminated, workspace runtime handles all transitions - **Workspace Purity Law (§16)**: Workspace Runtime retains navigation ownership — the Command Surface only expresses intent, never executes navigation - **Frontend Constitution (§17)**: The Command Surface continues to visualize reality (available actions) and surface next actions - **Constitutional Simplicity Rule (§4)**: No new abstractions, no new files, no new imports - **Canonical Ownership Law (§10)**: Navigation ownership confirmed as Workspace Runtime — no ambiguity introduced - **Universal Object Lifecycle (§7)**: Not affected — this change operates at the workspace navigation layer, not object lifecycle

Discovery Exit Criteria

Criterion	Status	Evidence
Canonical ownership confirmed	✓	Workspace Runtime (<code>store.ts</code>)
Every consumer identified	✓	1 consumer: <code>executive-home.tsx</code>
Every provider identified	✓	3 providers: react, store, api/client
Migration order known	✓	4 steps (see §12)
Rollback known	✓	Single git revert (see §13)
Verification defined	✓	14 test cases (see §14)
No unknown architectural dependencies remain	✓	All dependencies traced within CEP scope

Discovery Phase: Complete. Awaiting founder approval to proceed to Implementation.

16. Success Evidence (LX-06D-R3)

The measurable evidence that will prove this CEP succeeded after implementation:

#	Evidence	Measure	Verifiable By
1	Constitutional articles satisfied	Navigation Constitution §15 restored; no <code>window.location.href</code> in file	Static search: zero occurrences
2	Acceptance gates satisfied	§23 Gates 10, 17, 22 pass	Gate checklist
3	Canonical ownership established	Navigation intent owned exclusively by Workspace Runtime; Command Surface expresses intent only	Code review: no navigation execution in Command Surface
4	Founder experience metrics improved	Context preserved, cognition continuous, no mental restart, no page reload wait	Manual walkthrough of <code>/open</code> targets — verify no page reloads
5	Repository simplification achieved	0 files added; 0 files removed (constitutional fix); route map removed	<code>git diff --stat</code> : route map deleted, no new files
6	No regressions introduced	All existing Command Surface flows (creation, AI, voice, upload) still work	Full 321-test suite passes + manual flow checks
7	Command Surface consolidation enabled	CEP-002 unlocked (legacy command surface now safely removable)	CDR-002 readiness

17. Remaining Unknowns (LX-06D-R3)

No remaining architectural unknowns within CEP scope.

All dependencies within the approved CEP-001 scope have been traced and resolved: - Canonical owner: confirmed (Workspace Runtime) - Consumers: 1 identified (executive-home.tsx); false positive excluded (living-workspace imports its own command surface) - Providers: 3 identified (react, workspace store, api/client) - Route map targets: all 11 map to valid WorkspaceType values - Migration order, rollback, verification: defined

No unresolved architectural dependency remains within the CEP-001 scope. Implementation is constitutionally permitted pending founder approval.

CDR-001 produced by SHUNYA Constitutional Chief Architect Discovery constraints: Navigation Constitution §15, Workspace Runtime boundaries, executive-home command surface scope only Conforms to LX-06D-R2 (Discovery Gate) and LX-06D-R3 (Success Evidence + Remaining Unknowns)